

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2013-14

Zoology

Paper: ZOB-601 — Functional Anatomy and Economic Importance of Chordata

Time: Three hours

Full Marks: 70

Note: Answer six questions including Question No. 1 which is compulsory. Draw diagrams wherever necessary. The figures in the right-hand margin indicate marks.

Question 1. Answer the following parts:

() State whether the following statements are True or False, giving reasons in 2-3 sentences:
[4 × 2 = 8]

- (i) Ostracoderms were primitive jawed vertebrates.
- (ii) Heterocoelous vertebra possesses convexities on both side.
- (iii) Osteoblast cells form dentine layer of the tooth.
- (iv) Spiral valve in the intestine is found in shark fishes.

() Match the terms of Column-A with Column-B: [6 × $\frac{1}{2}$ = 3]

Column-A

- (i) Pallium
- (ii) Autostylic
- (iii) Omasum
- (iv) Ctenoid
- (v) Arcualia
- (vi) Neuromasts

Column-B

- (1) Stomach
- (2) Cerebral hemisphere
- (3) Vertebra
- (4) Rheoreceptors
- (5) Jaw
- (6) Ceruminous glands
- (7) Scale

() Differentiate between the following: [3 × 2 = 6]

- (i) Mullerian and Wolffian ducts
- (ii) Afferent and Efferent branchial arteries
- (iii) Nails and Hoofs.

() Define the following: [3 × 1 = 3]

- (i) Antivenin
- (ii) Dipnoi
- (iii) Holobranch.

Question 2. Describe the different types of jaw suspensorium in vertebrates. [10]

Question 3. Explain the following:

[2 × 5 = 10]

- () Types of uterus
- () Poultry.

Question 4. Give an account of various types of feathers found in birds.

[10]

Question 5. Discuss the changes that have occurred in cerebral hemisphere from fishes to mammals.

[10]

Question 6. Explain the following:

[2 × 5 = 10]

- () Dentition
- () Urinary bladder.

Question 7. Give an illustrated account of morphological and anatomical changes that have occurred during the course of vertebrate origin.

[10]

Question 8. Write notes on the following:

[2 × 5 = 10]

- () Accessory respiratory organs in fishes
- () Role of amphibians in biological control.

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